



KZCN

Zirconia-based Composite

KEY PROPERTIES

- High chemical stability
- High corrosion resistance
- Good thermal shock resistance
- Good wear resistance
- Stable at high temperature

APPLICATIONS

- Furnace parts
- Melting crucibles
- Friction wear parts

MAIN PROPERTIES

Property	Value	Unit
Density	3.5	g/cm ³
Fracture Toughness	3.0	MPa·√m
Young Modulus	205	GPa
Hardness	11	GPa

* All properties measured at 20°C unless otherwise stated.

** Measured in a wall thickness of 2 mm

OPERATING CONDITIONS

Max working temp (air)	1500 °C
Recomended temp	<1400 °C
Continuous temp	1300 °C

ELECTRICAL BEHAVIOR

Insulator

MATERIAL INFORMATION

Material type	Zirconia-based composite
Structure	Multi-phase ceramic
Key feature	Chemical stability
Machinability	Medium
Composition	Al ₂ O ₃ (50%) / ZrO ₂ (30%) / SiO ₂ (<20%)

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

The information provided is for guidance only and does not constitute a guarantee.

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